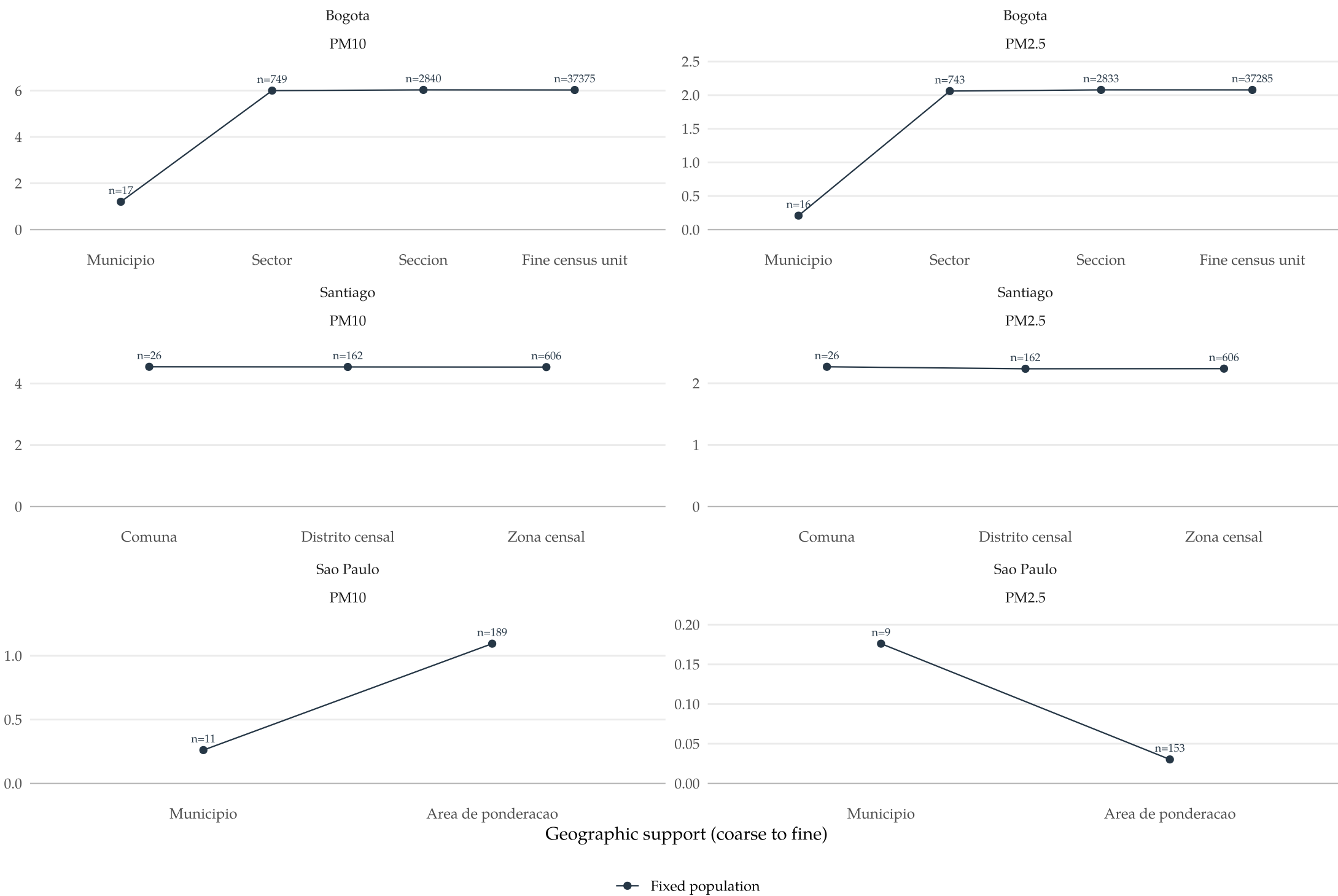


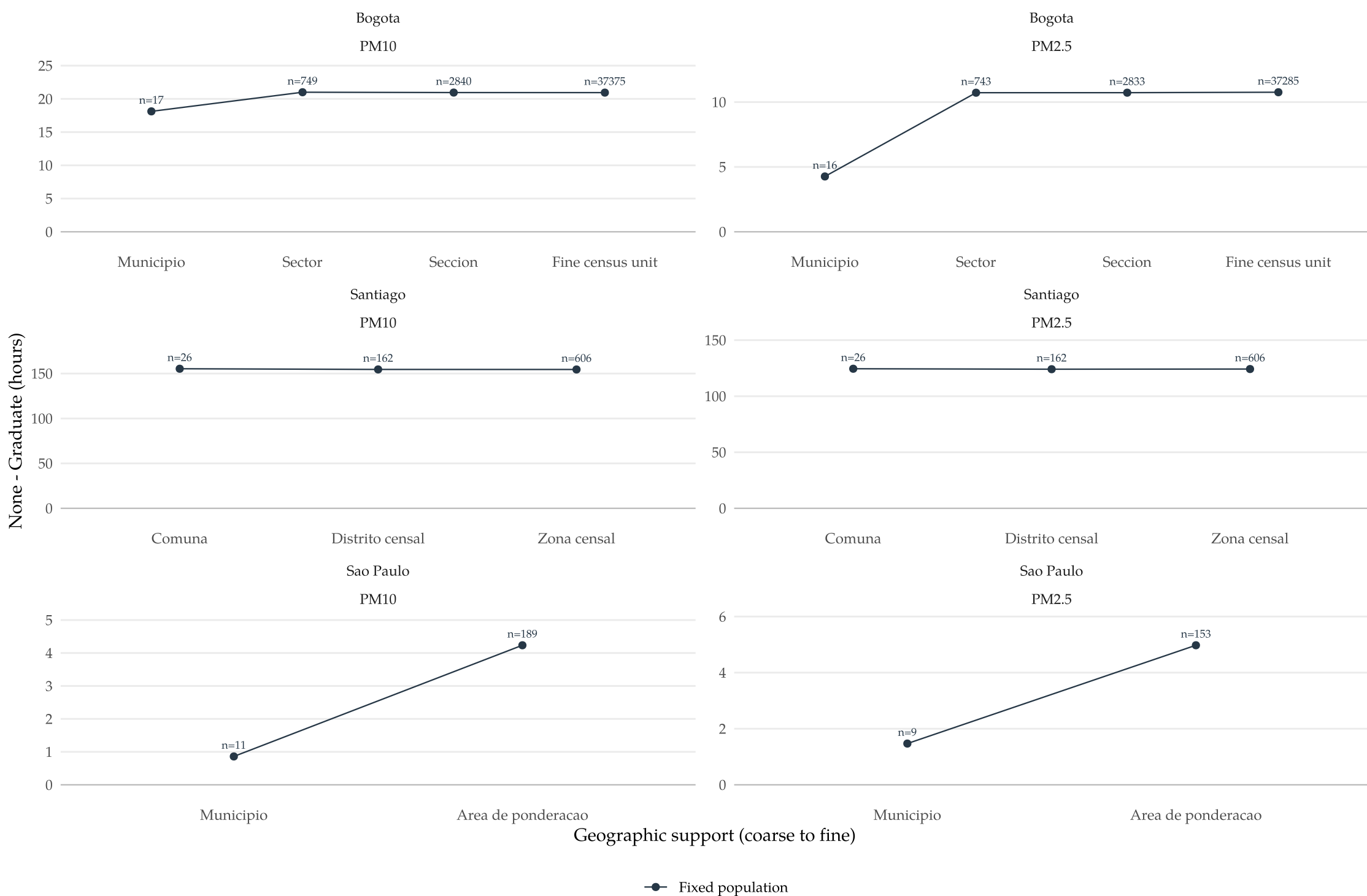
Design A - annual mean exposure

None - Graduate (micrograms per cubic metre)



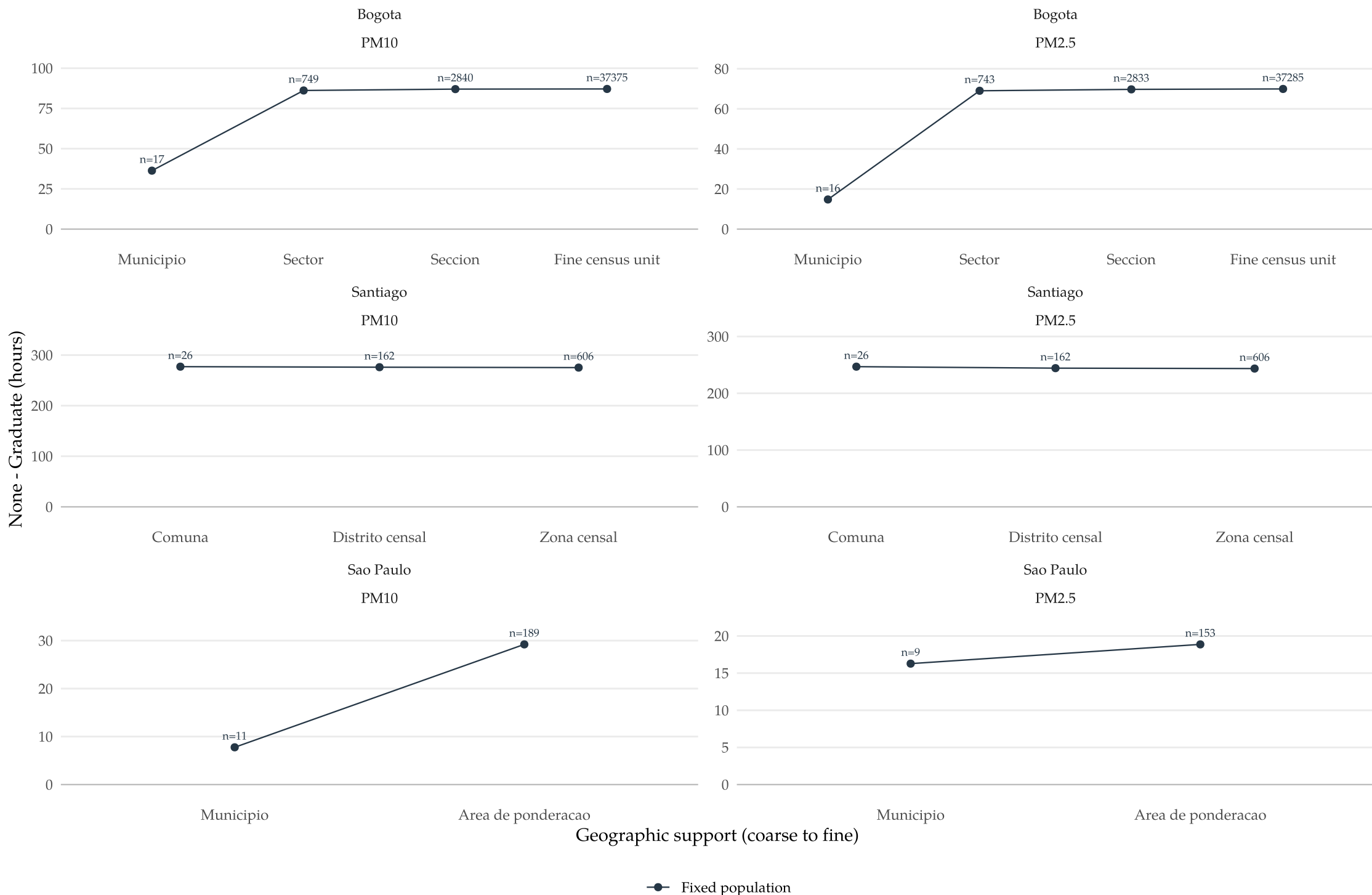
Fixed education-reporting adults and individual levels. Exposure aggregated with all-adult weights; locality branch shown separately.

Design A - annual hours at or above IT1



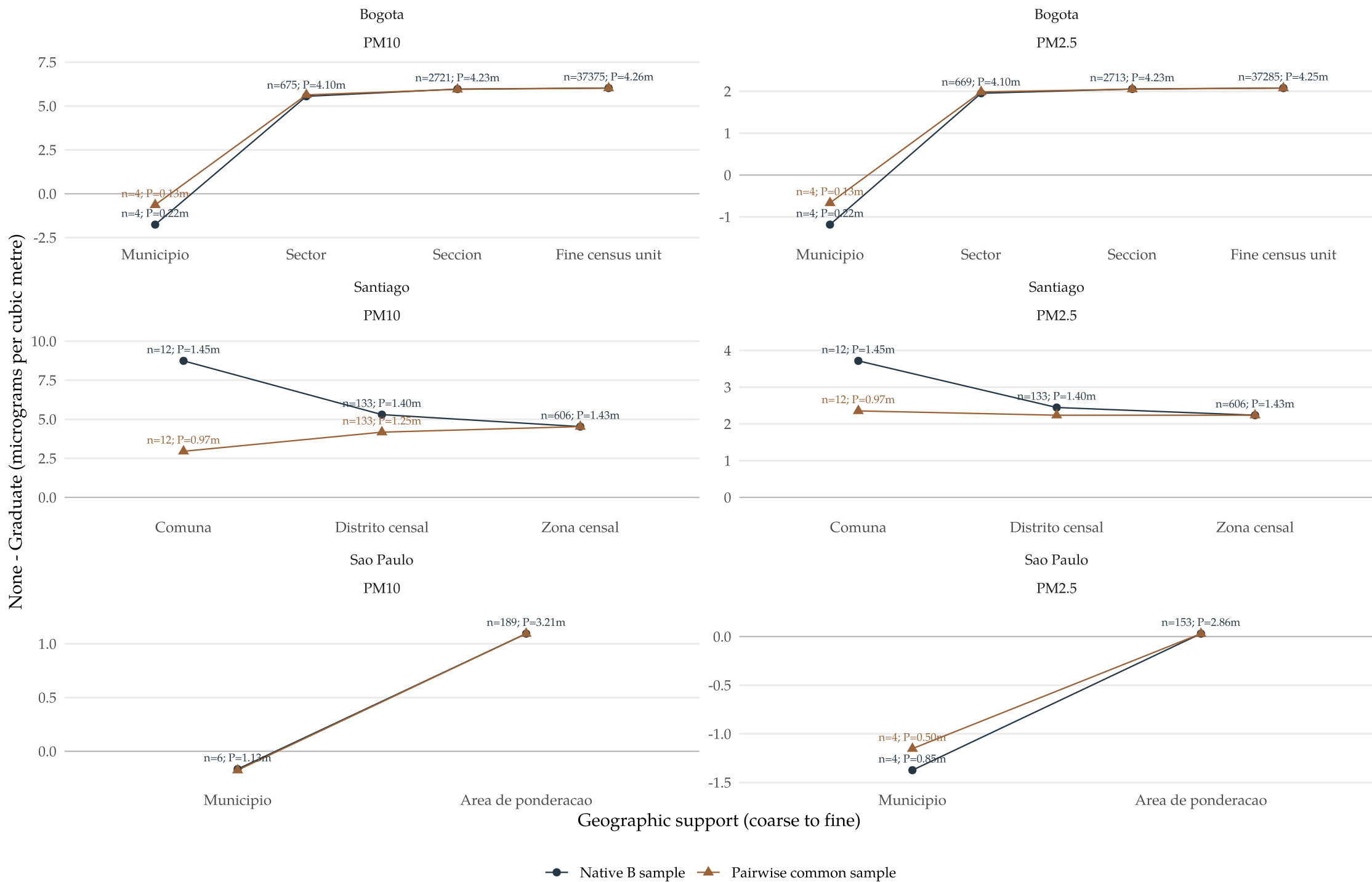
Fixed education-reporting adults and individual levels. Exposure aggregated with all-adult weights; locality branch shown separately.

Design A - annual hours at or above IT2



Fixed education-reporting adults and individual levels. Exposure aggregated with all-adult weights; locality branch shown separately.

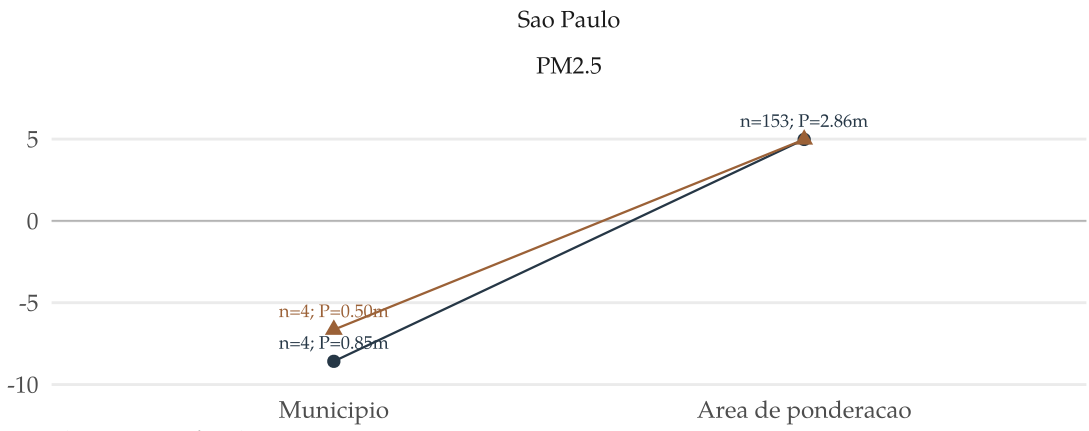
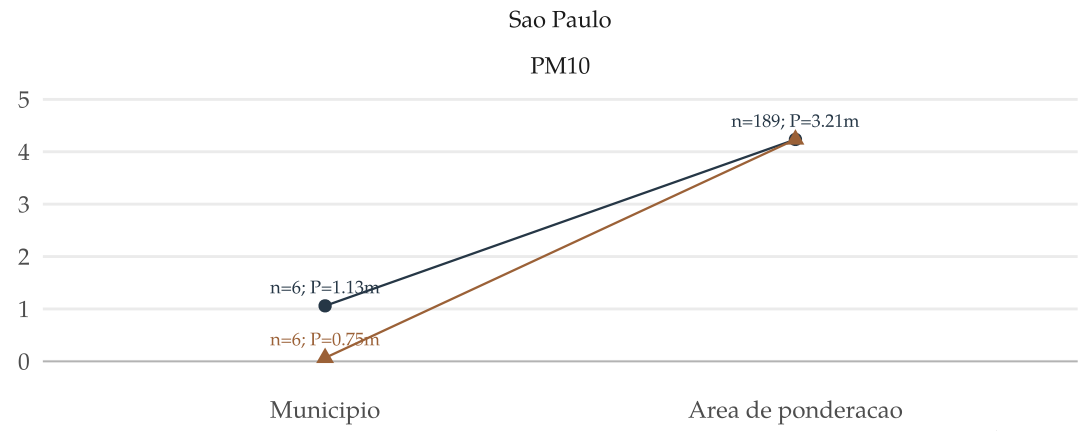
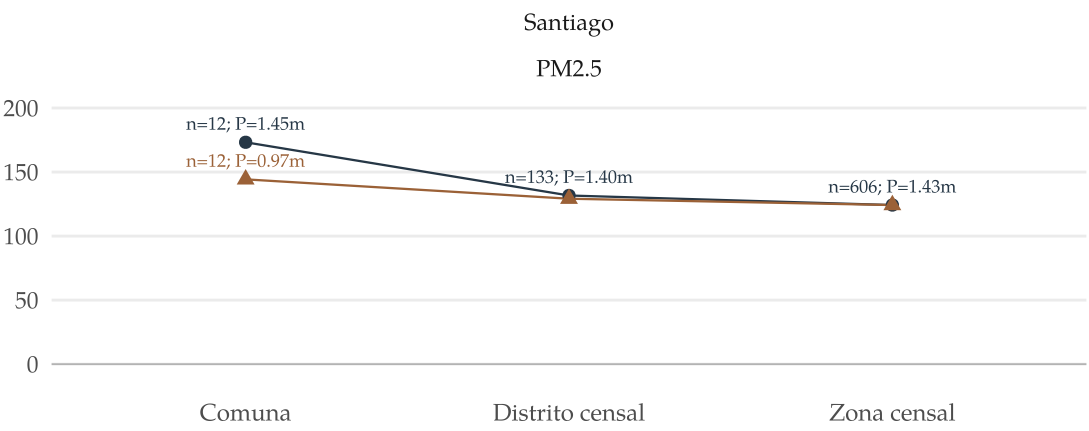
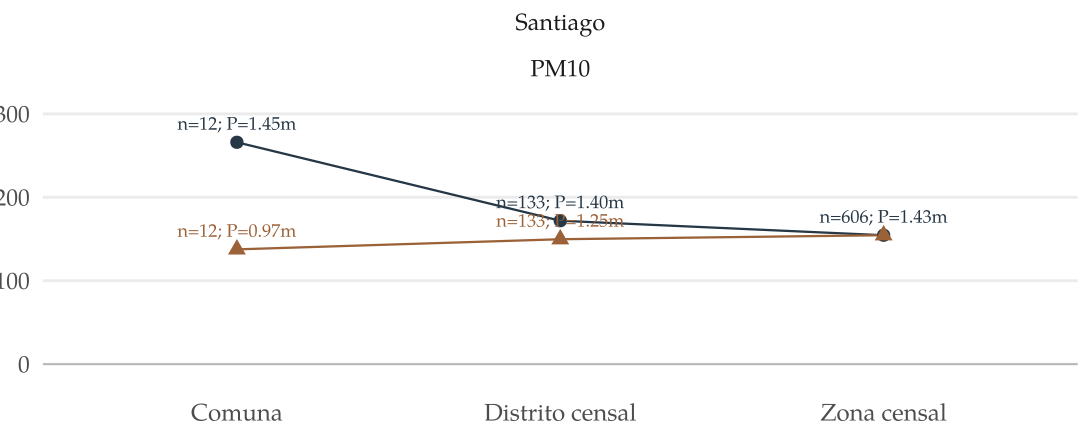
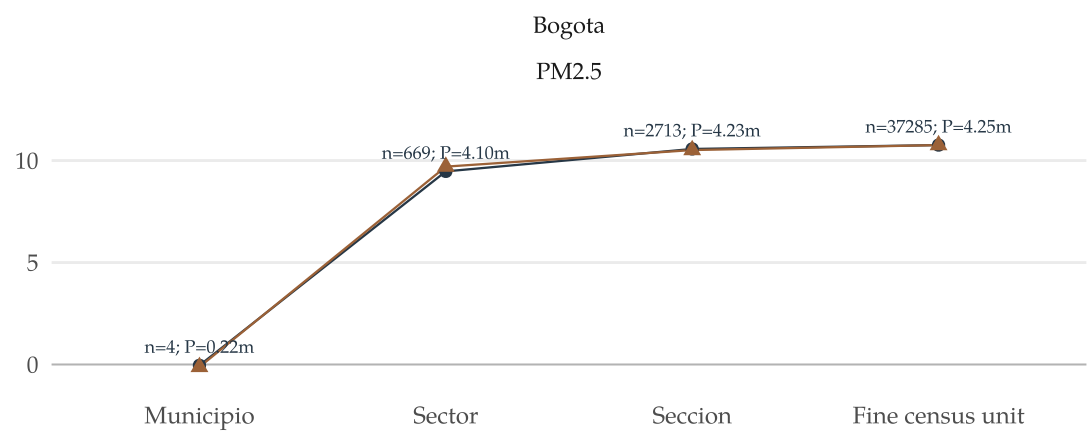
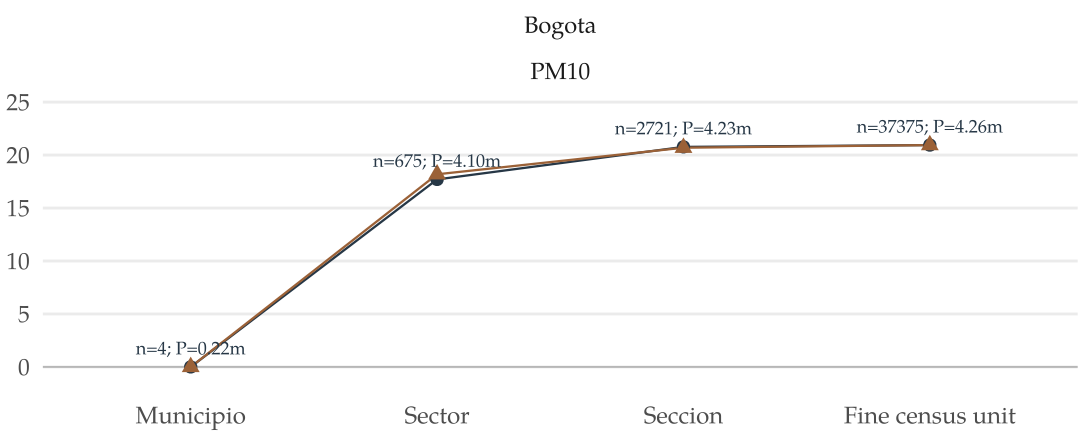
Design B - annual mean exposure



Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

Design B - annual hours at or above IT1

None - Graduate (hours)

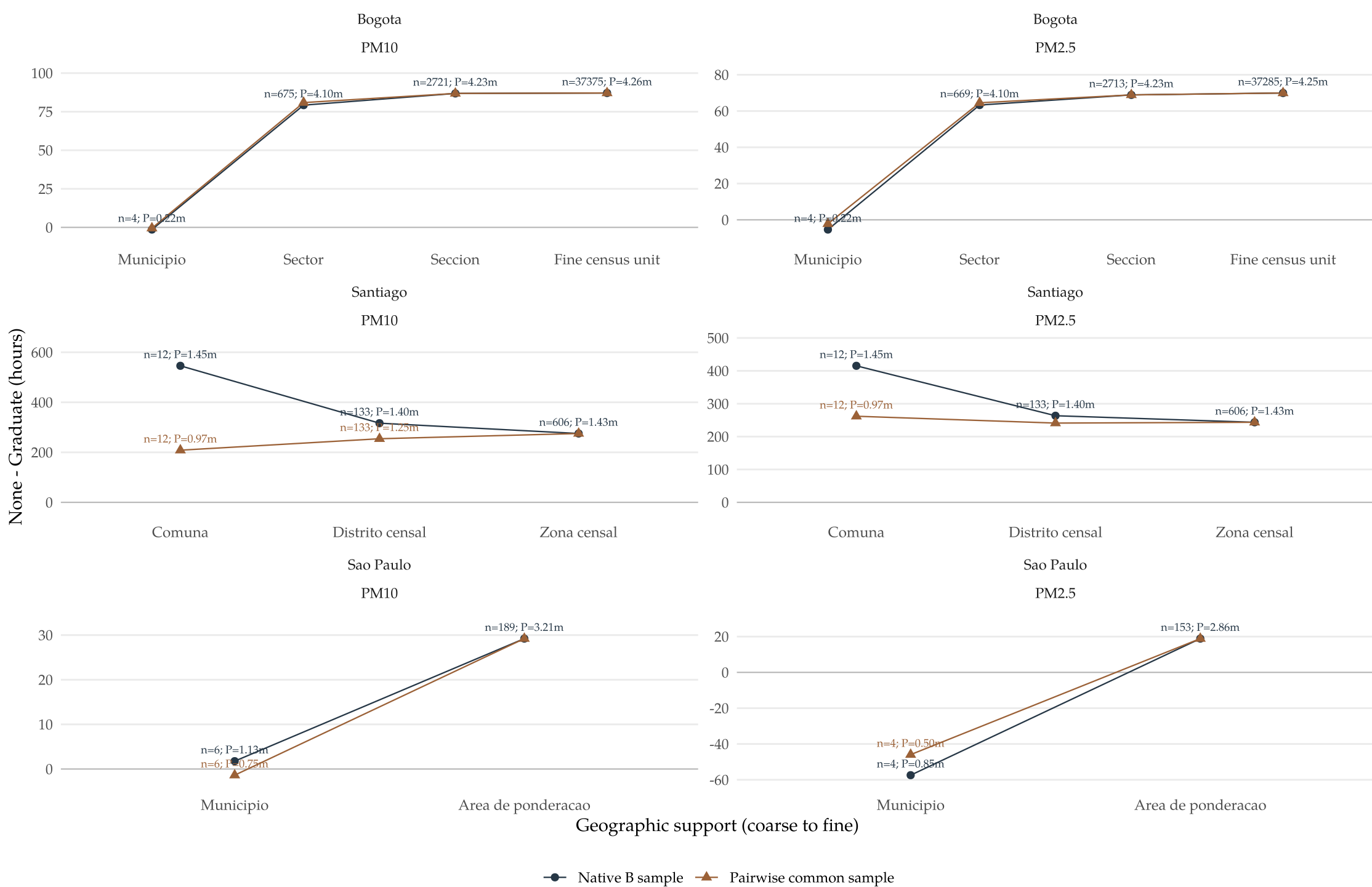


Geographic support (coarse to fine)

Native B sample Pairwise common sample

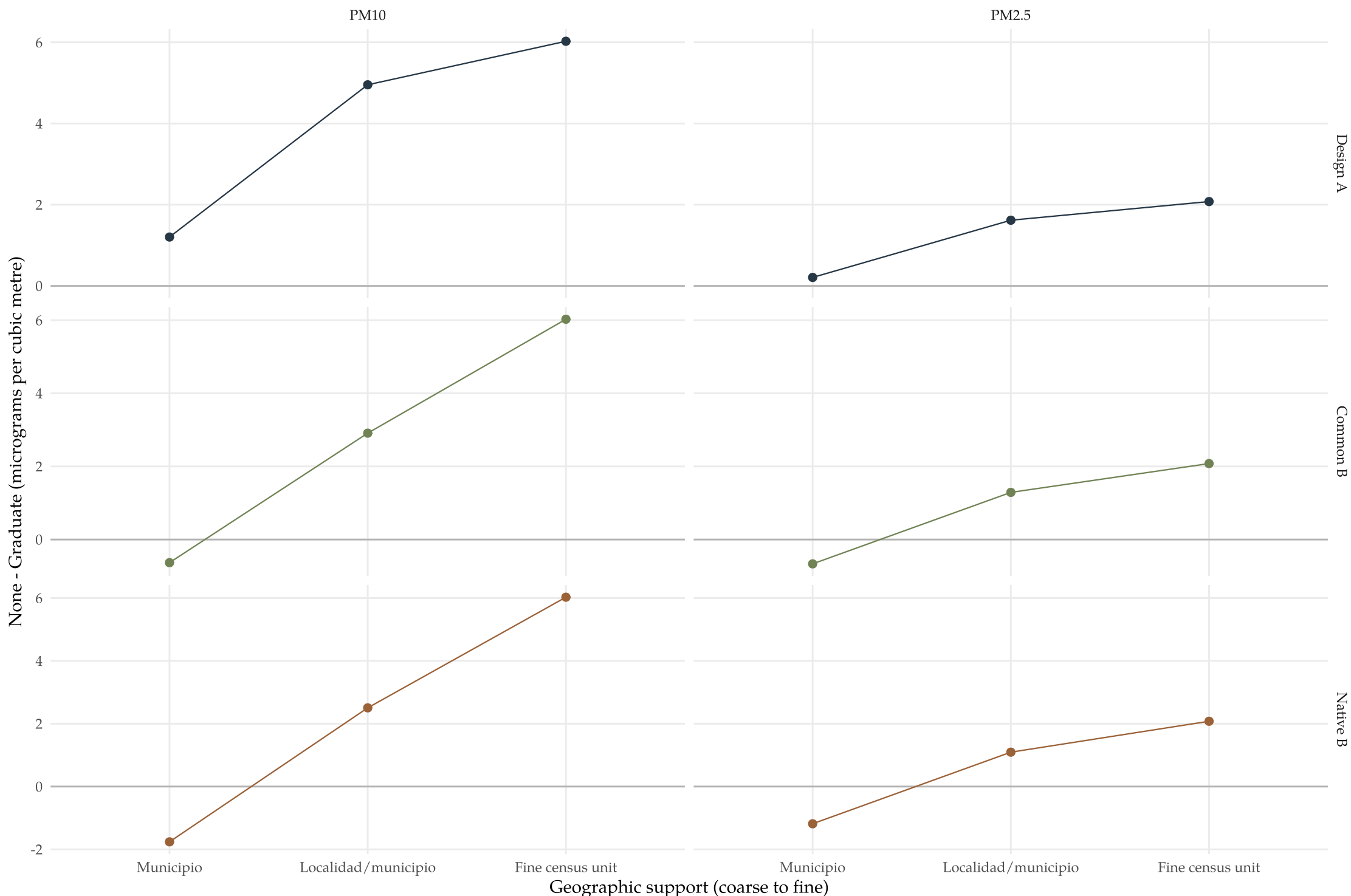
Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

Design B - annual hours at or above IT2



Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

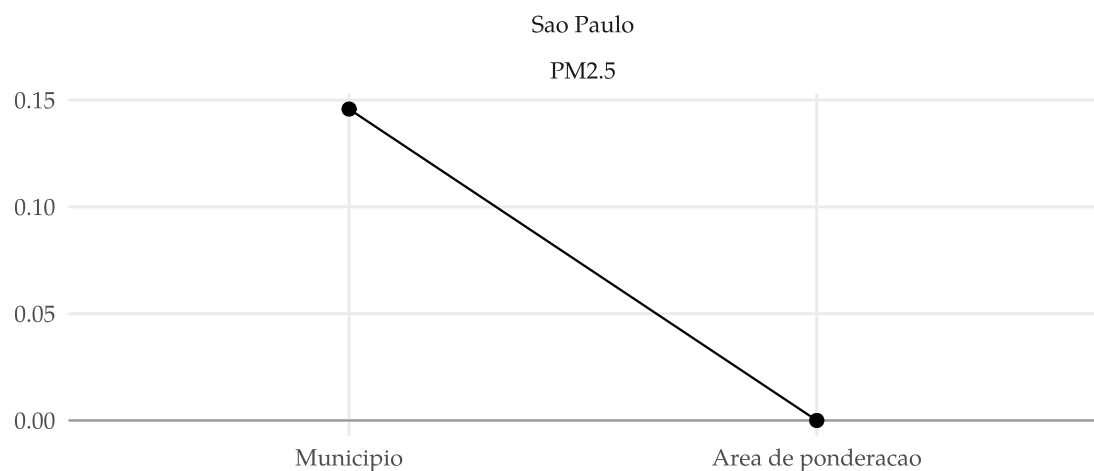
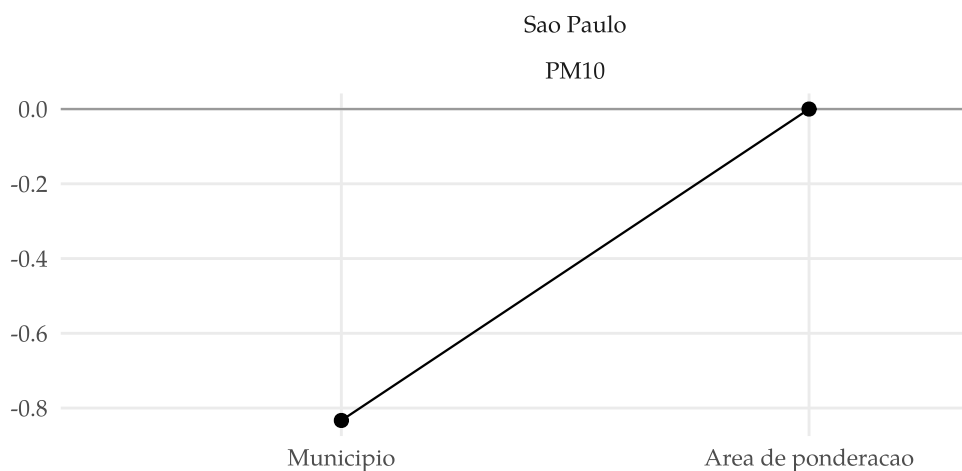
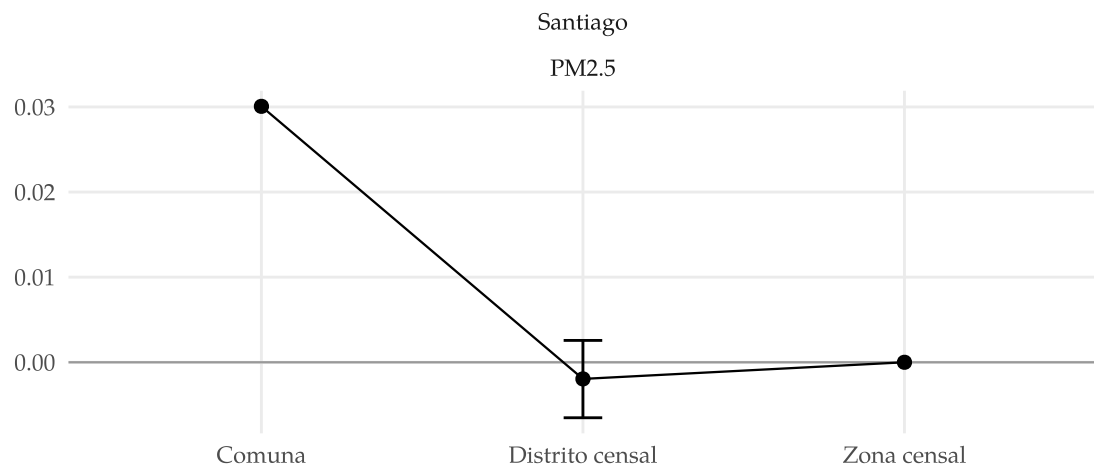
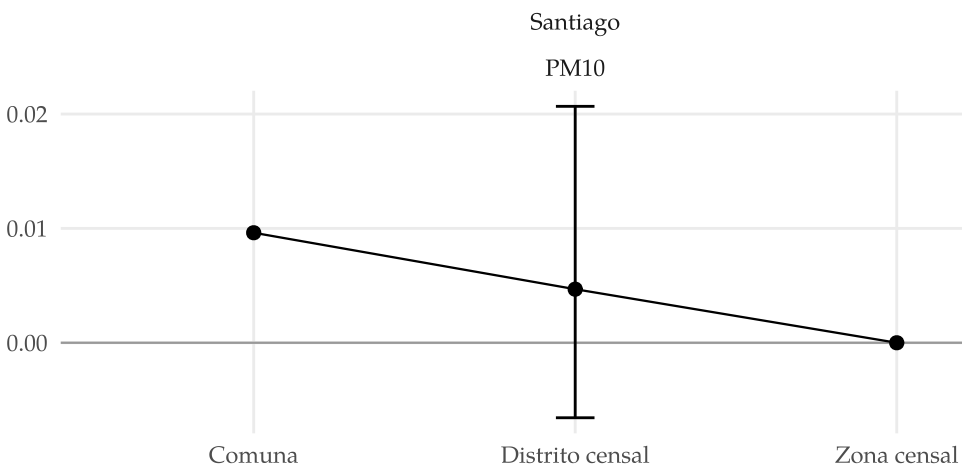
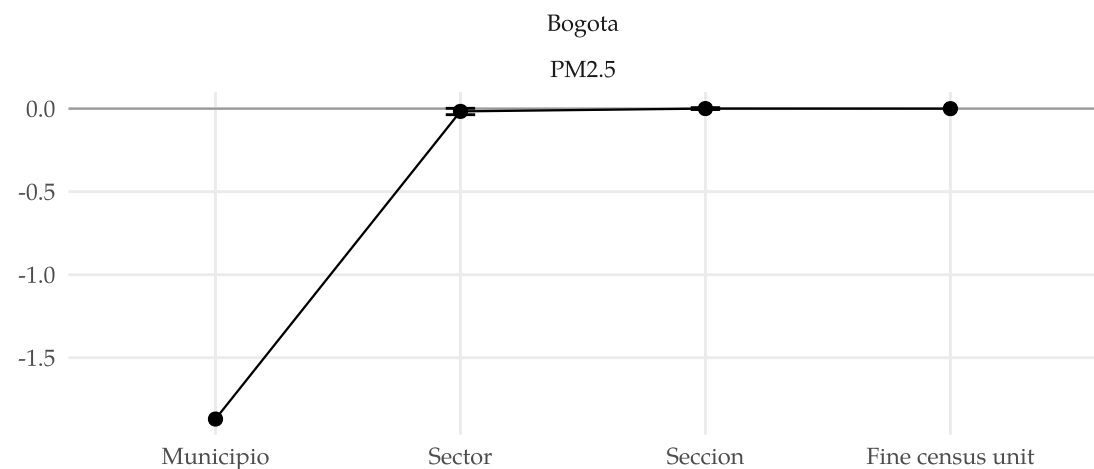
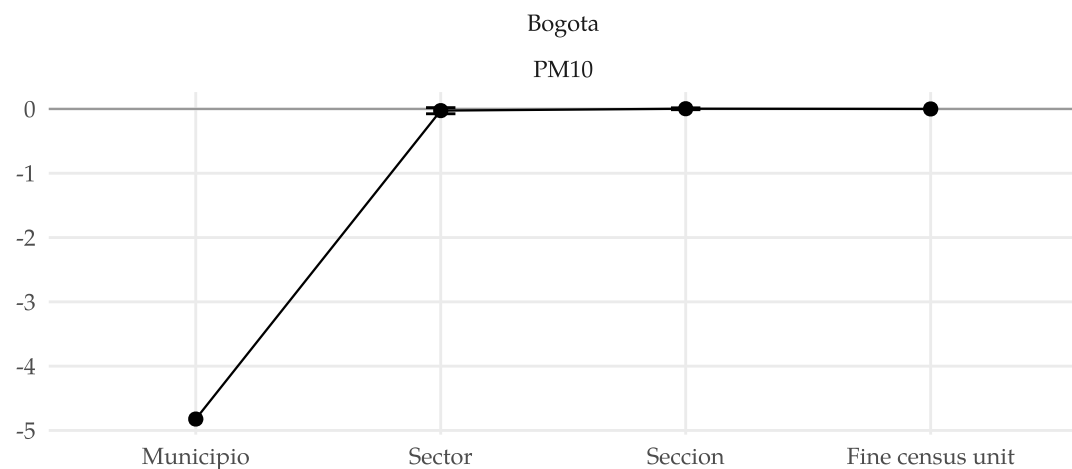
Bogota: separate locality branch



Descriptive comparisons.

Design A - change relative to finest support

Change in absolute None - Graduate gap

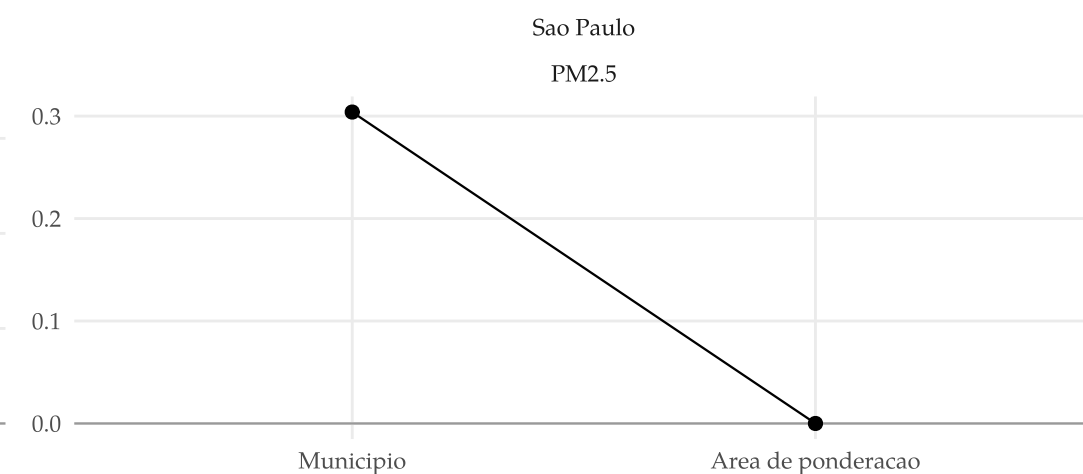
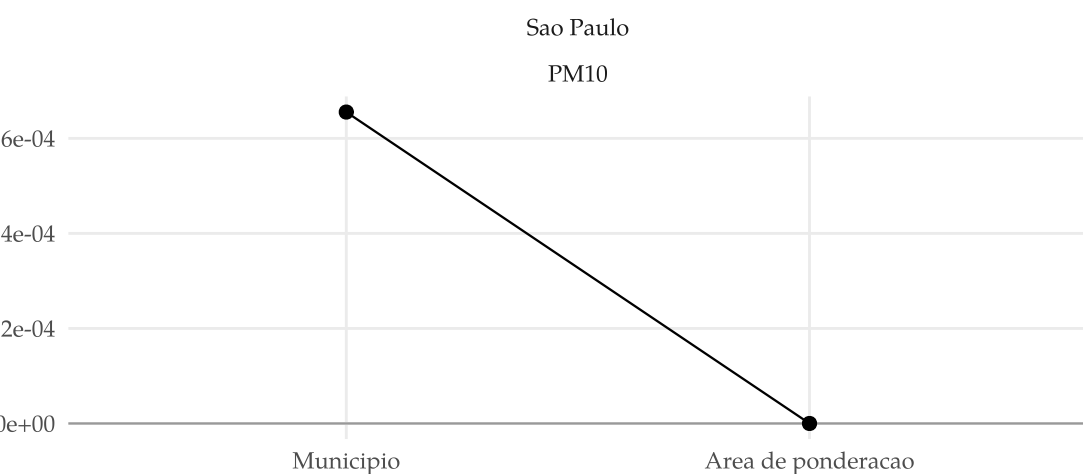
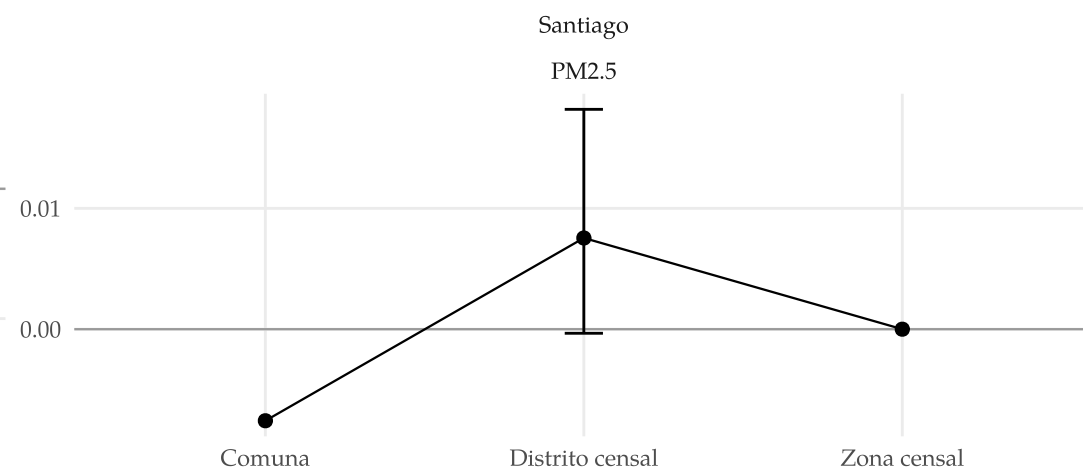
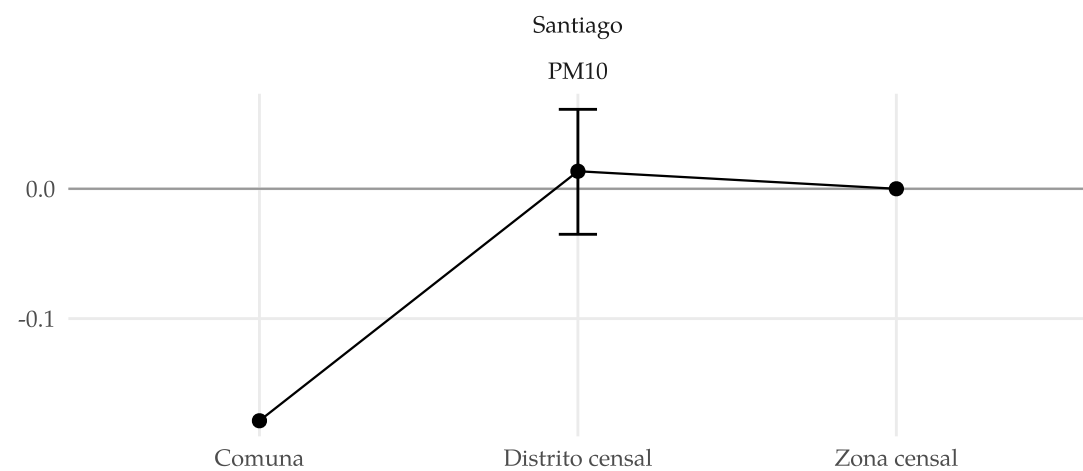
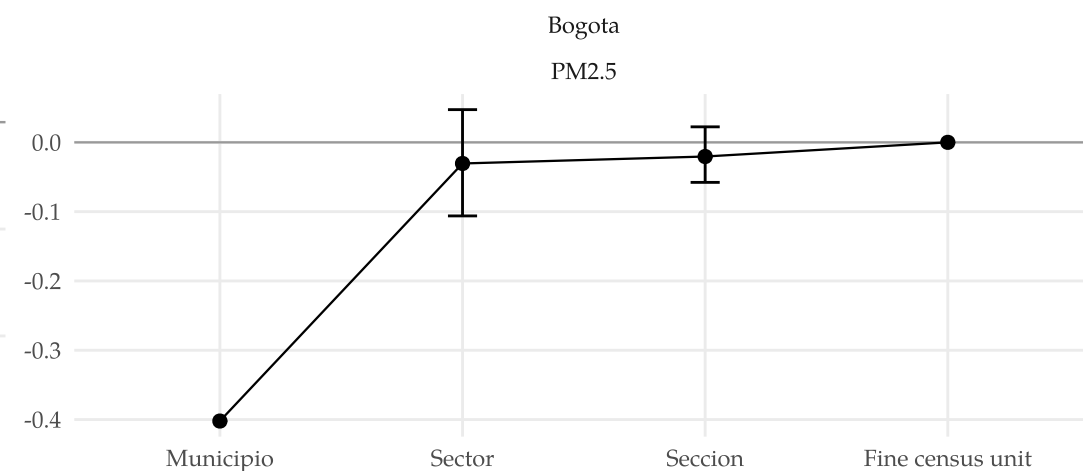
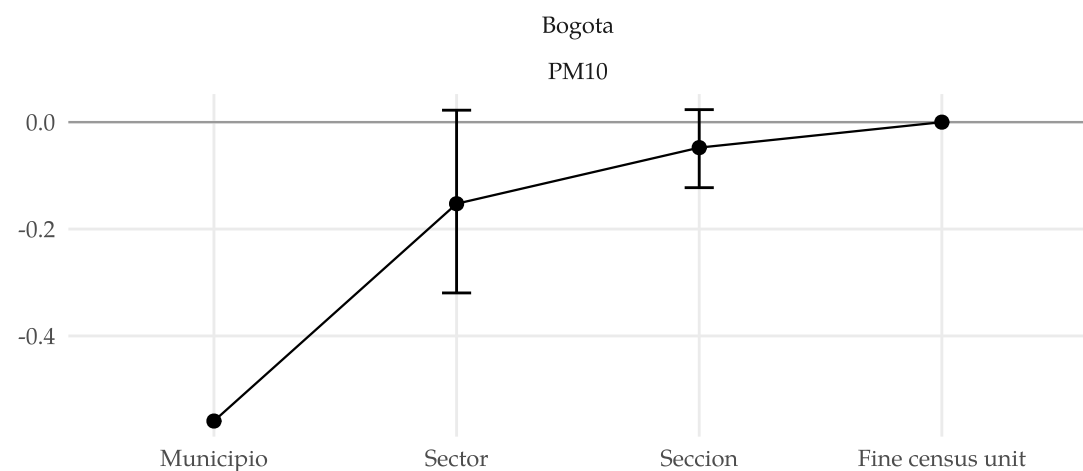


Geographic support (coarse to fine)

95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.

Common-sample B - change relative to finest support

Change in absolute None - Graduate gap



Geographic support (coarse to fine)

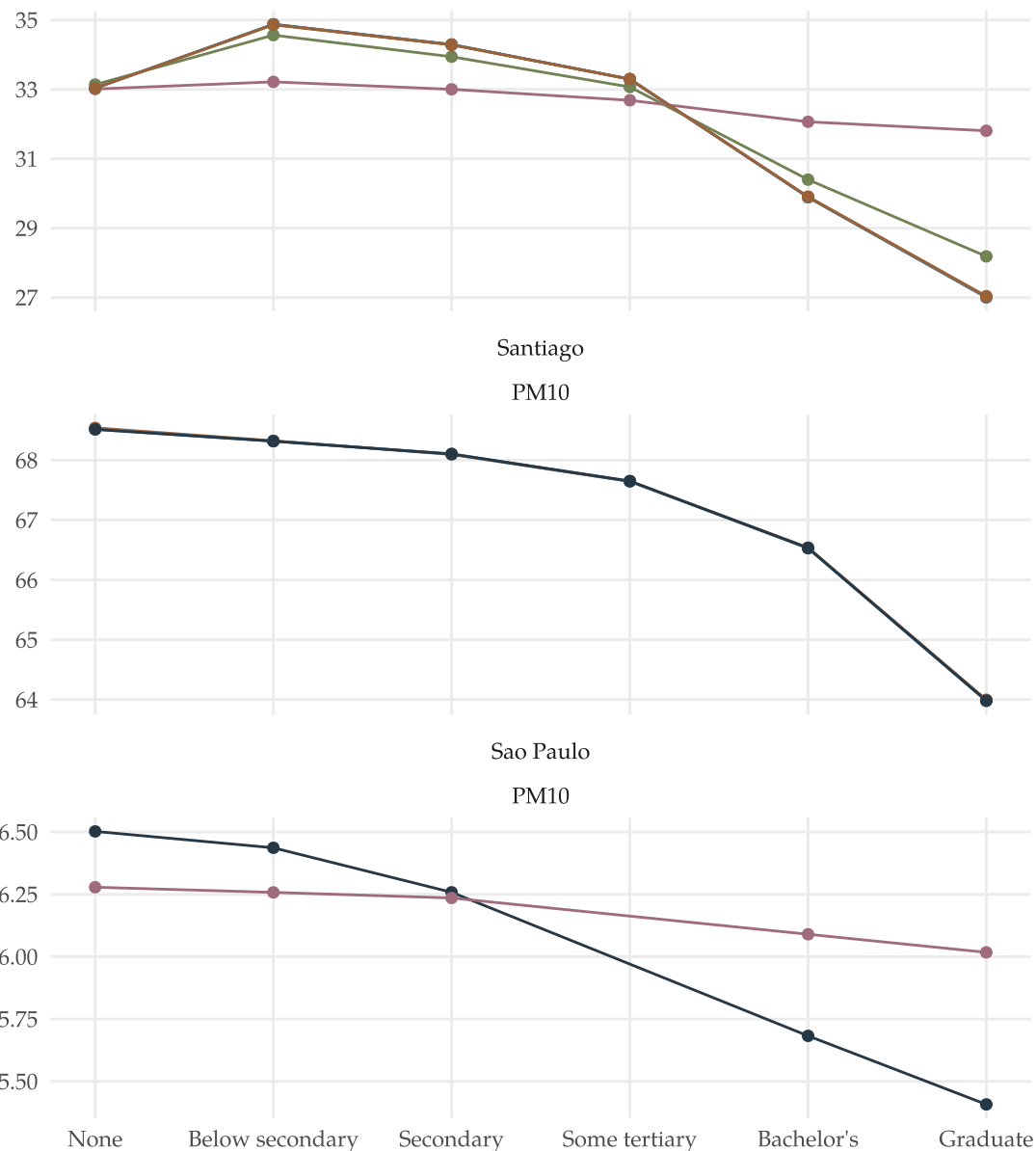
95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.

A - full education-group profiles

Mean exposure (micrograms per cubic metre)

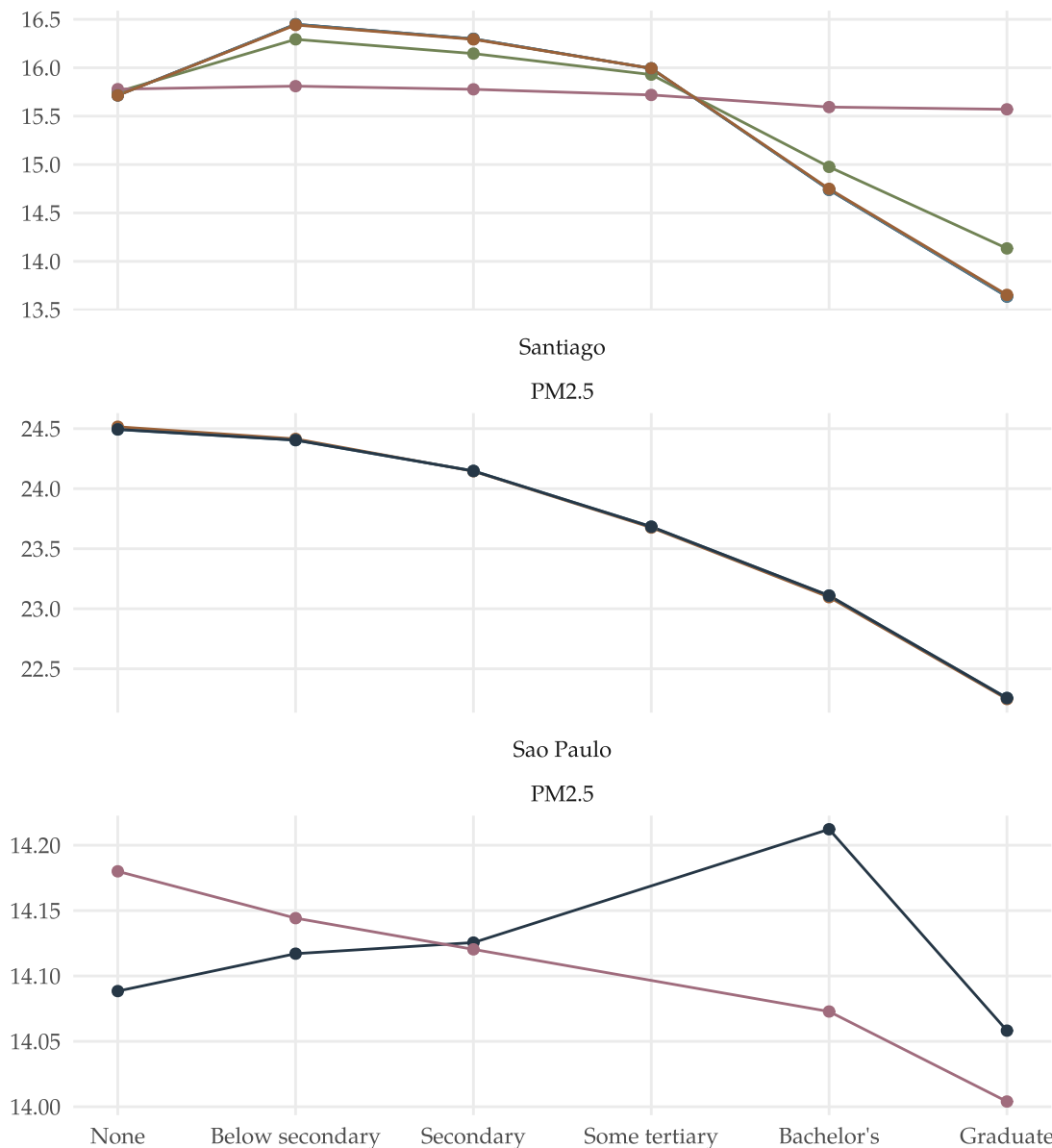
Bogota

PM10



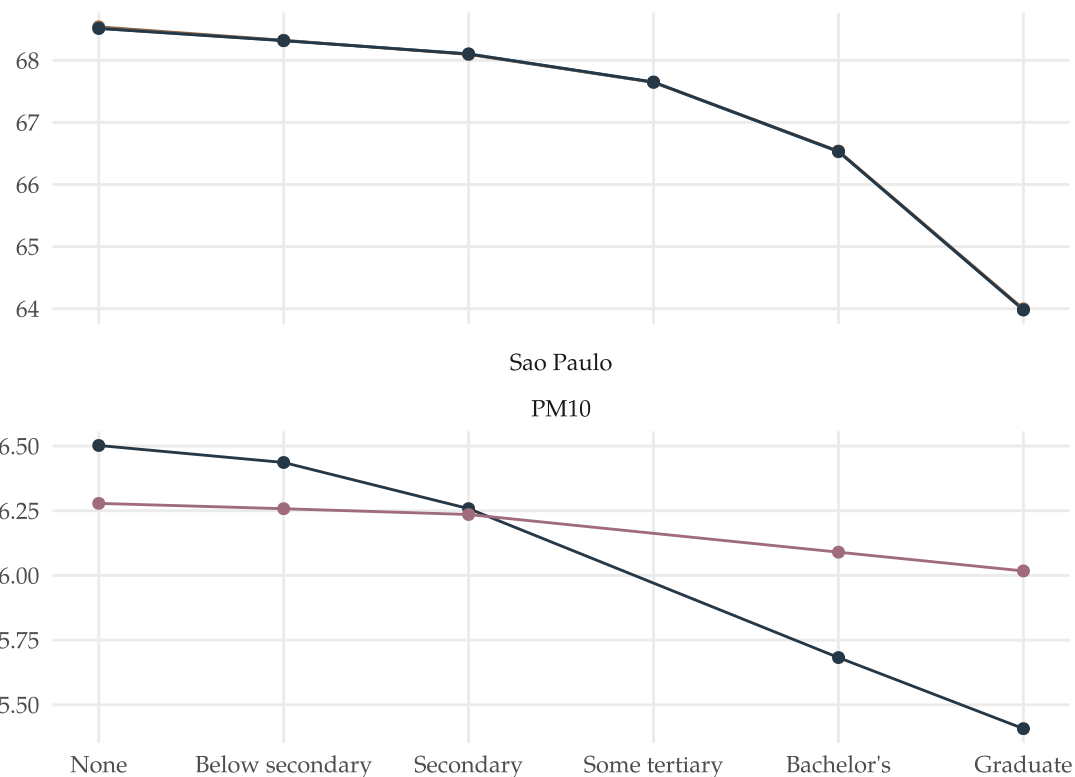
Bogota

PM2.5



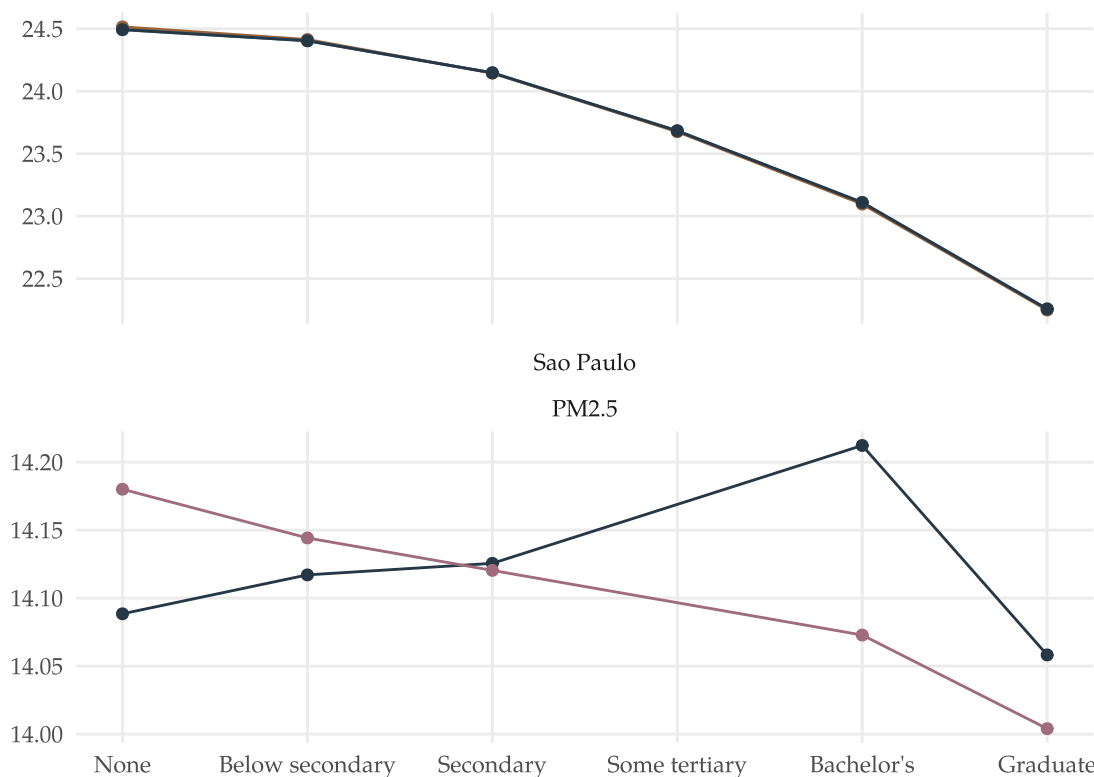
Santiago

PM10



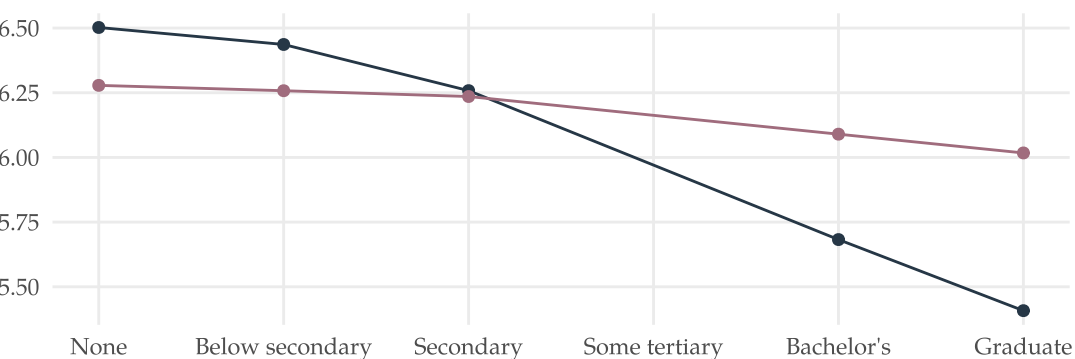
Santiago

PM2.5



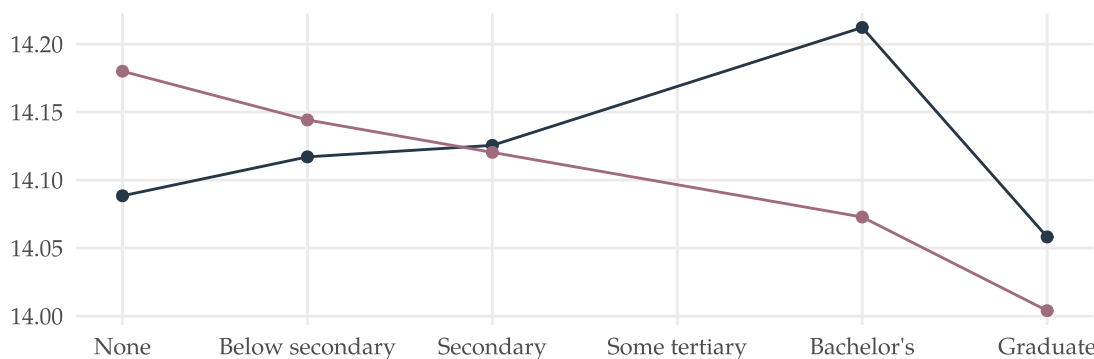
Sao Paulo

PM10



Sao Paulo

PM2.5

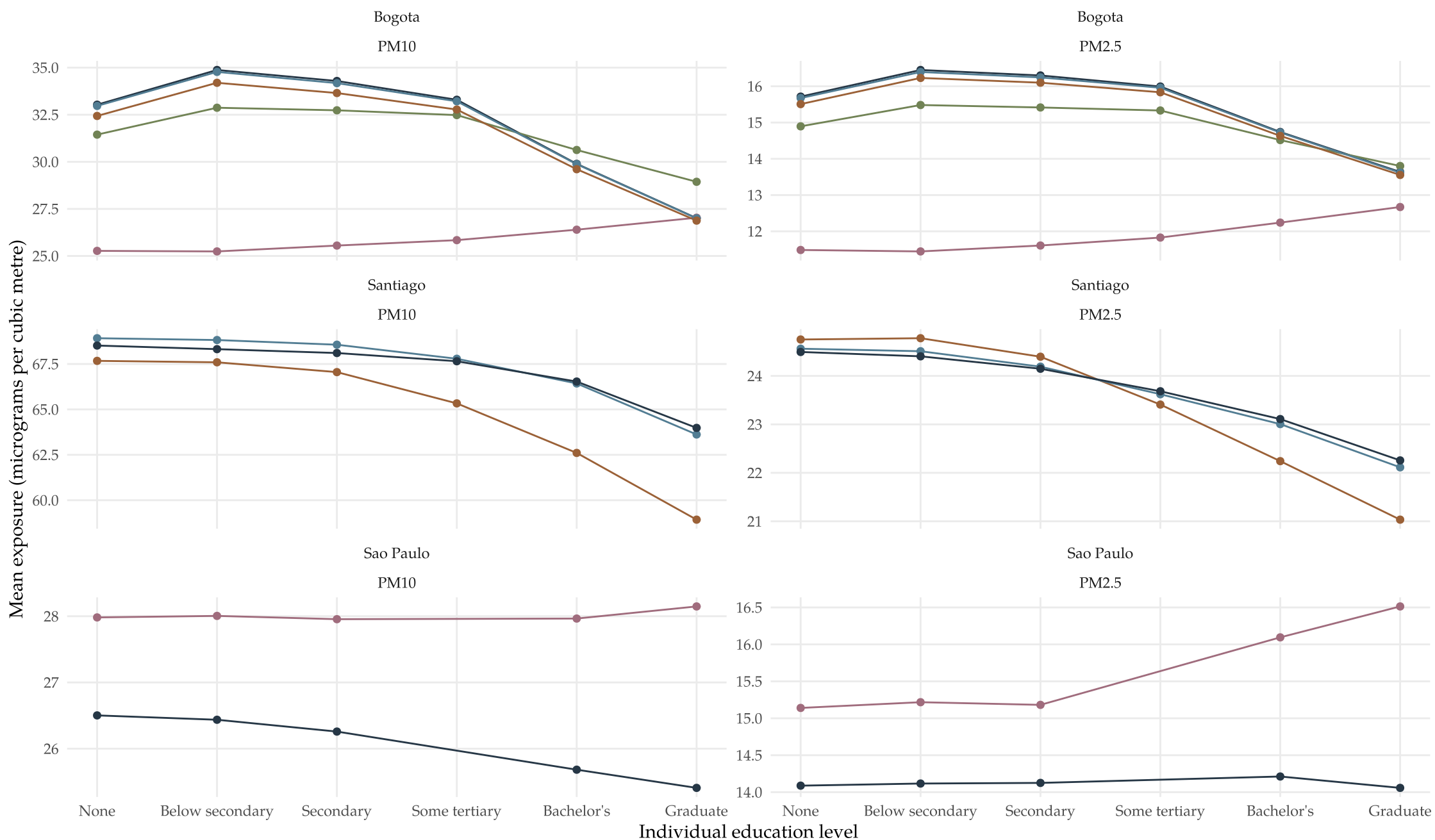


Geographic support

- Area de ponderacao
- Distrito censal
- Localidad/municipio
- Seccion
- Zona censal
- Comuna
- Fine census unit
- Municipio
- Sector

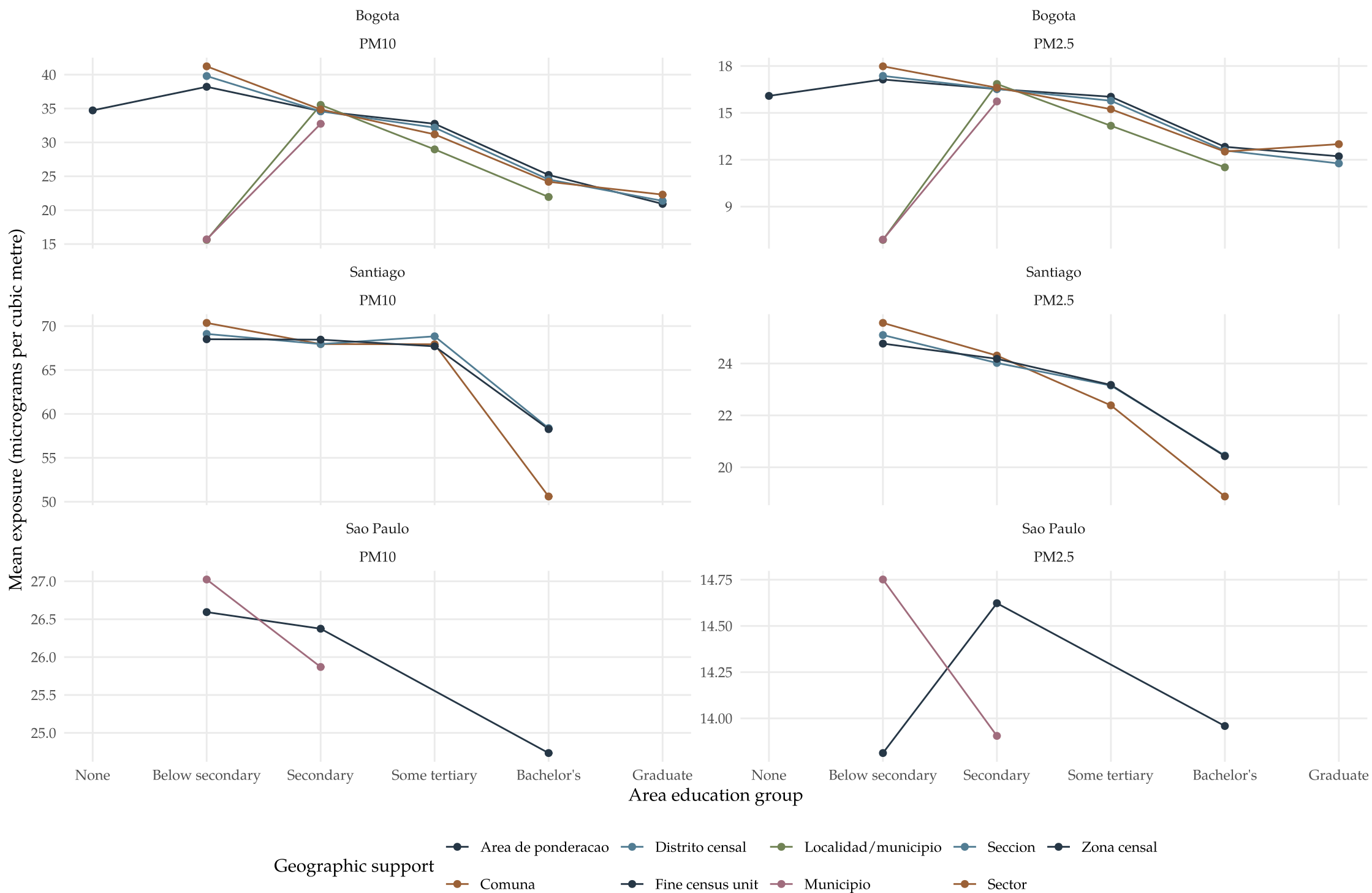
Harmonized education levels; group means use existing person weights. Sao Paulo's census cannot code 'Some tertiary'. Bogota's 'Some tertiary' includes completed tecnica and normalista.

Native B - full education-group profiles



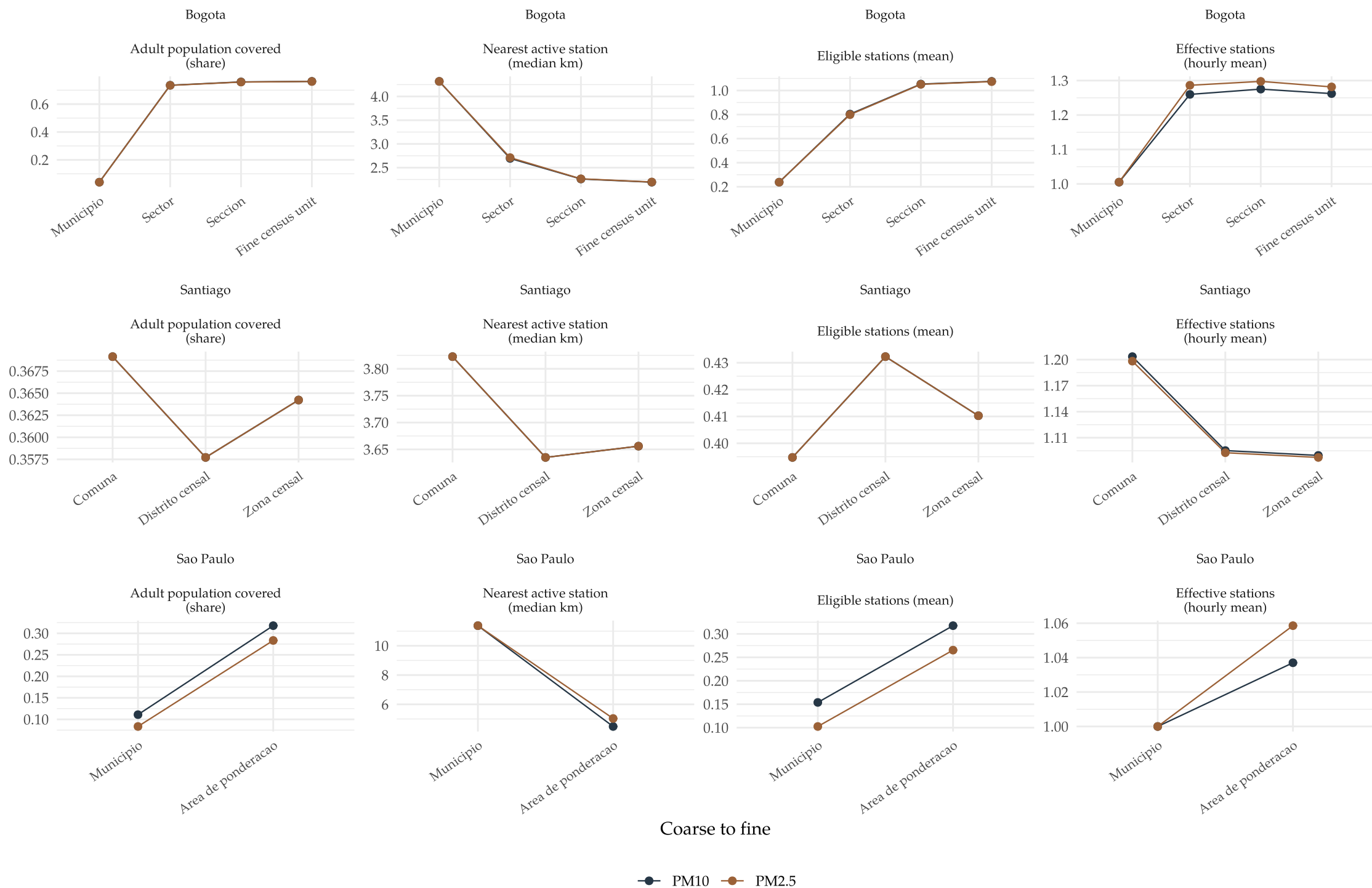
Harmonized education levels; group means use existing person weights. Sao Paulo's census cannot code 'Some tertiary'. Bogota's 'Some tertiary' includes completed tecnica and normalista.

C - full education-group profiles



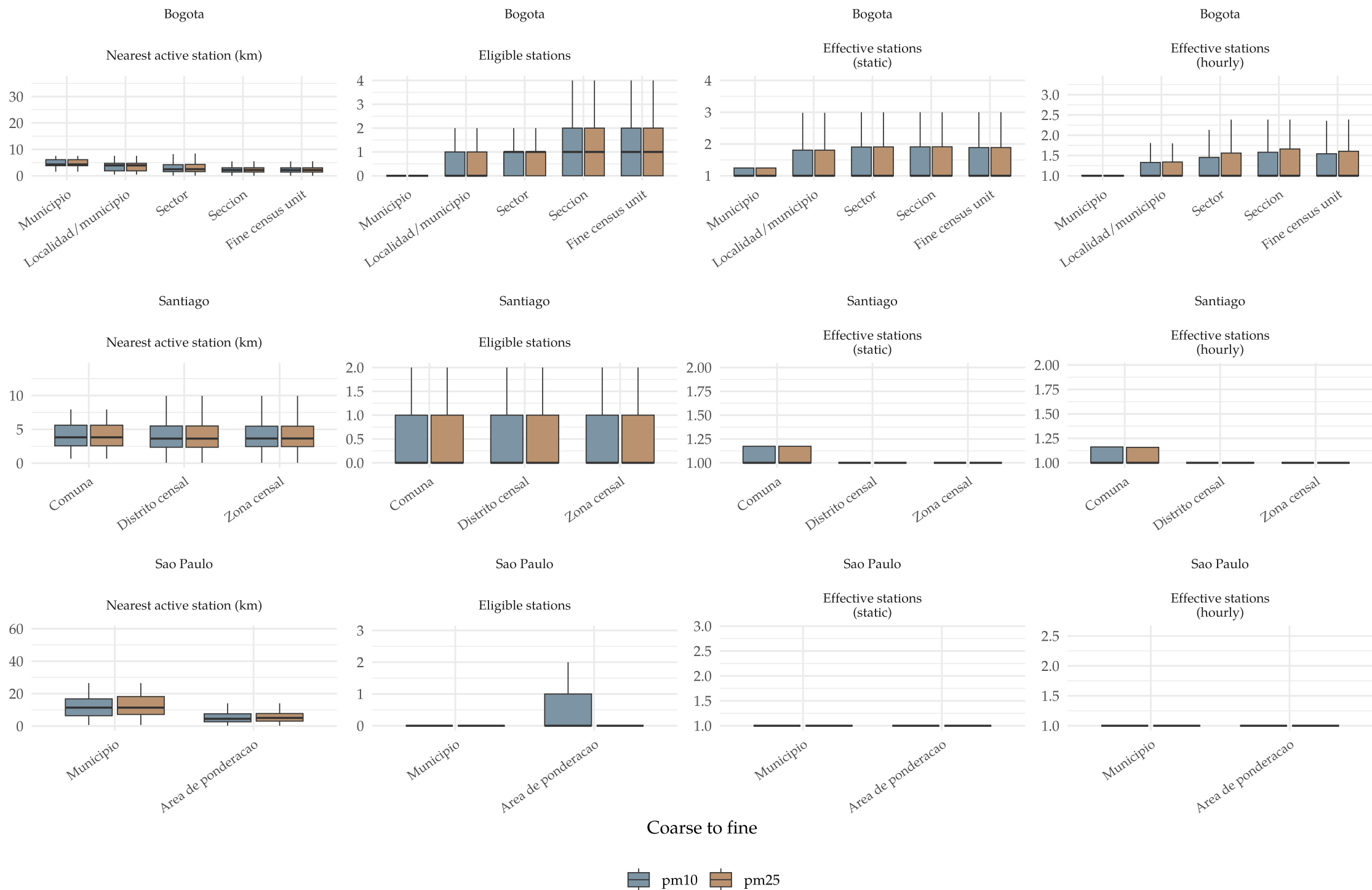
Classification only; finest exposure fixed. Empty area groups remain unavailable. These groups differ from A/B levels.

Monitoring-matrix sensitivity under Design B



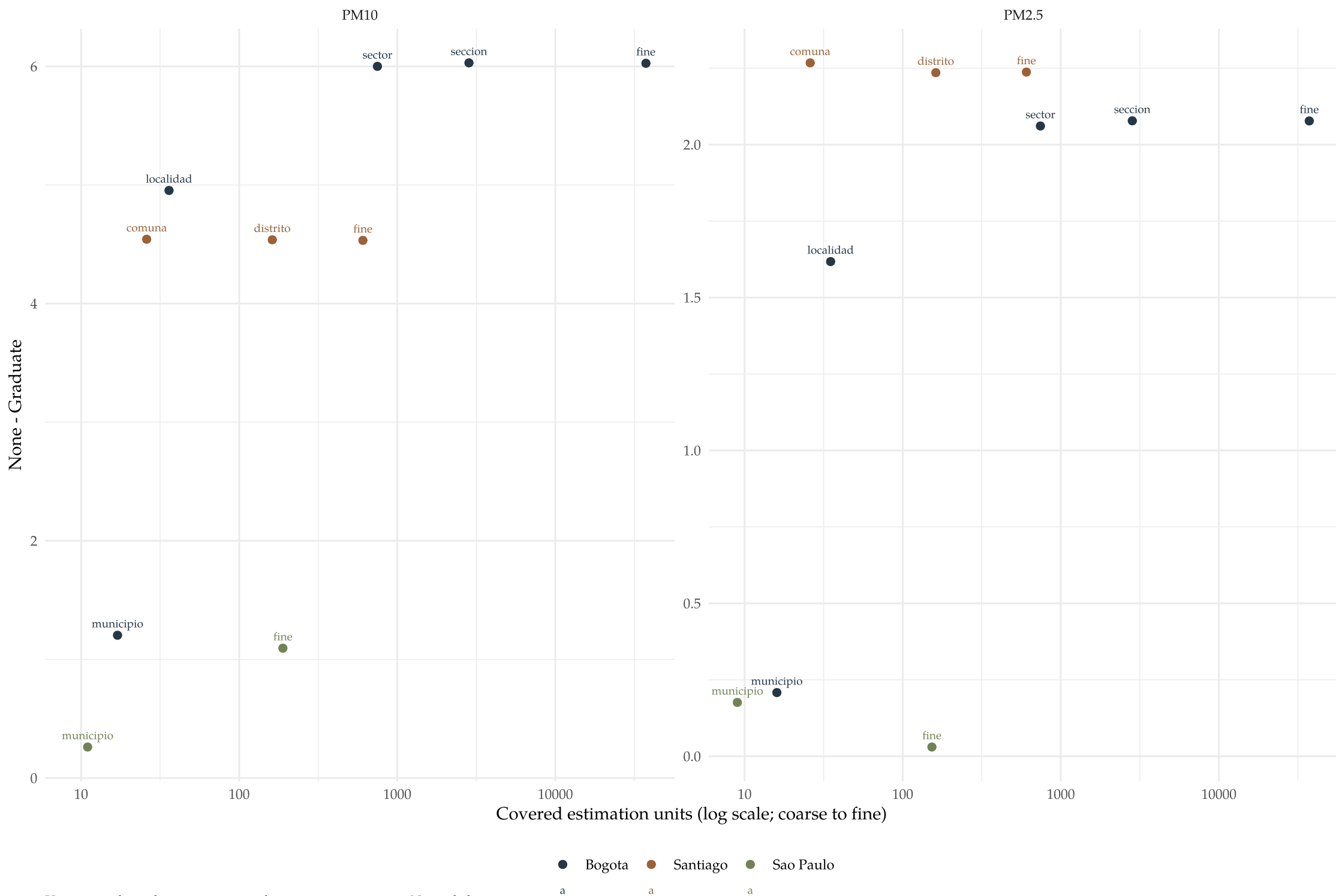
Catalog, active, eligible and contributing station counts are distinct in the tables.

Monitoring-matrix distributions across populated units



Boxes: unweighted unit quartiles; whiskers: 1.5 IQR. Outliers omitted from view only.

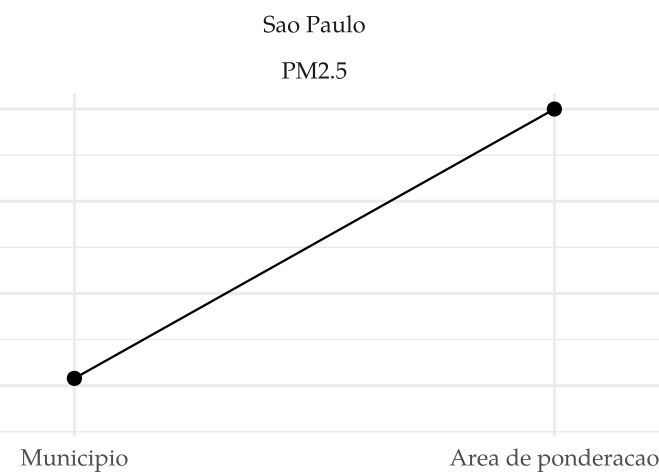
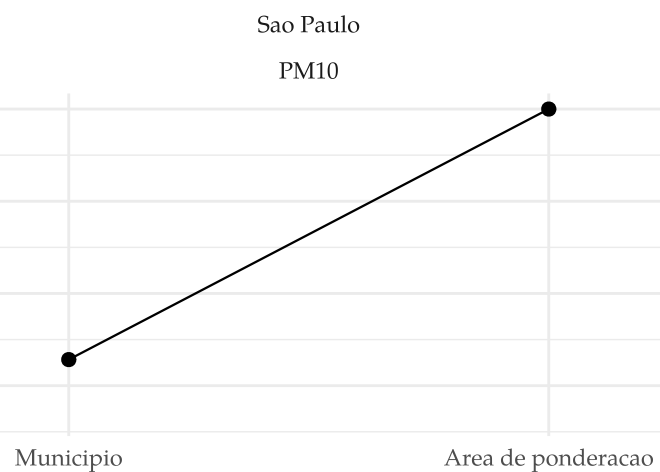
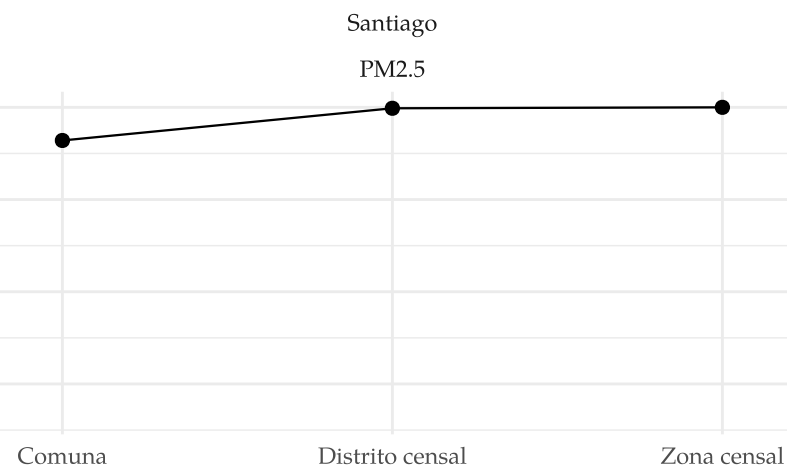
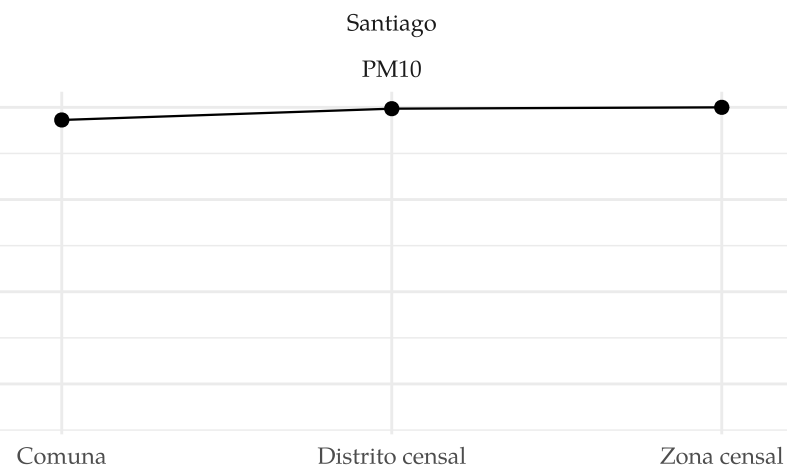
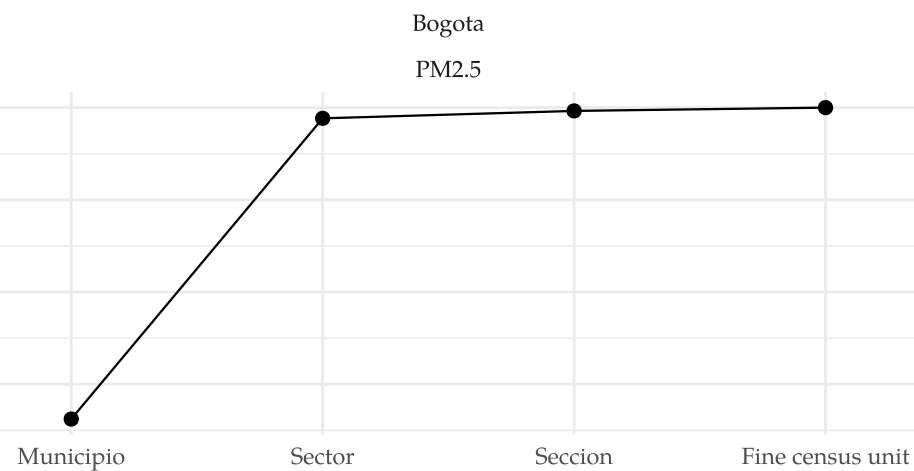
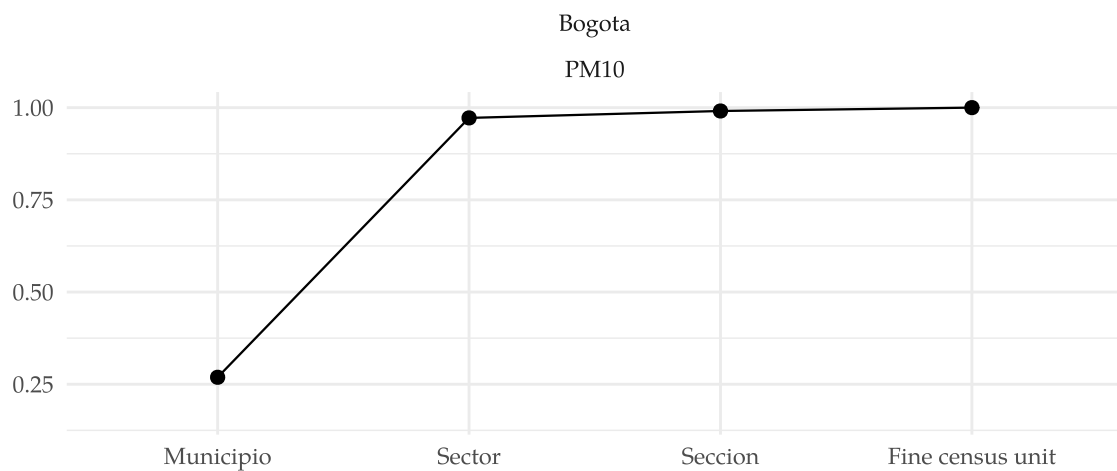
Design A: covered-unit counts as a supplementary resolution measure



Unit counts do not harmonize geographic concepts across cities. No pooled estimate.

Design A: exposure variance retained

Share of finest-level variance



Coarse to fine

All-adult aggregation weights and fixed covered support. Declining variance does not imply a monotone gap.